Two types of transmission are

i. Primary transmission and
ii. Secondary transmission

1.4.1 Primary transmission

In the primary transmission, the voltage produced from power plants, transmitted over the transmission lines to the substations with the help of step up transformer.

1.4.2 Secondary transmission

The voltage transmitted by the primary transmission is reduced by a step down transformer, and is transmitted to the sub-stations in the city.

1.4.3 Advantages of AC transmission

i. Alternate current can produce voltage (33 KV) at the highest level.
ii. The voltage can be increased or decreased.
iii. High voltage transmission reduces losses.
iv. It is easy to maintain sub-stations and less expensive.

1.4.4 Disadvantages of AC transmission

i. The AC line has the capactive and inductive effect.
ii. Due to skin effect, it requires more effective conductors.
iii. The construction of AC transmission is complicated.
iv. More copper conductors are required for transmitting AC.

1.5 Overhead line transmission

The overhead lines are used to transmit electricity from power plants to consumers. A network of electrical power lines are used to transmit the electricity over a geographic area is called power grid.

1.5.1 Main components used in overhead lines

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i. Electric conductors
ii. Poles and towers
iii. Insulators
iv. Cross arm that holds electrical insulation materials
v. Transformer, lightning arrester, blocking barriers to the pole, and circuit breakers

Do you Know?



The voltage occurs when the solar light on the system with two semi conductors. Photons are absorbed in solar light when sunlight shines in one semi-conductor. Thus the electrons are energized and move to another semi-conductor. A small voltage is produced between two semi-conductors. This is called Solar cell.

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